

Tidal: Direction-Paired Virtual-Channel Buffer Pooling with a 1R1W Shared Pool

Changxun Pan Liming Wang

Equal contribution: 50% each

Abstract

Conventional on-chip routers bind virtual-channel (VC) storage to individual input directions. This organization strands storage when one direction is congested and its opposite peer is idle. TIDAL reorganizes the same eight VC slots per opposing-direction pair into two two-slot reserved regions and a four-slot shared POOL. Each direction retains one private adaptive VC and one private escape VC. The POOL provides only one read and one write service per cycle; alternating read phases let one side inspect its reserved region while the other inspects the POOL. Four conserved POOL credits control admission. A pressure policy moves released credits toward the side with fuller reserved storage, keeps the previous owner on ties, caps ownership at three, and reclaims only idle peer-owned slots on demand.

We implement the design in gem5 Garnet and evaluate an equal-storage four-private-VC baseline. Across 651 main-matrix simulations, all runs completed without deadlock and recorded zero cross-direction POOL read conflicts. Under persistent 90/10 X-direction skew with eight-cycle links, TIDAL lowers network latency by 9.2% and extends the coarse stable offered rate from 0.08 to 0.10. At a fully skewed point beyond baseline saturation it delivers 1.818 times as many packets. Under tornado traffic, pressure-aware credit control lowers latency by 4.85%, whereas blind round-robin ownership loses 56.5% throughput. Balanced, short-link, and rapidly reversing traffic can be slower. The results identify persistent directional skew near an input credit limit as the useful operating region and motivate timing and RTL cost validation.

Keywords: on-chip networks, virtual channels, shared buffers, credit flow control, Garnet, gem5

1 Introduction

Virtual-channel flow control improves throughput by multiplexing logical channels over physical links, but it also makes buffer organization a central router design choice [3]. Garnet models a pipelined, credit-controlled network inside gem5 and is widely used to study these tradeoffs [1, 2]. Its conventional private organization is simple: input-direction VCs cannot migrate. When traffic is directionally skewed, a busy side can therefore stall while unused capacity remains stranded at its opposite peer.

Fully shared buffers improve utilization, but unconstrained sharing requires multiported storage or arbitration that can erase the intended cost benefit. Our question is narrower: *can an equal-capacity, direction-paired buffer recover useful elasticity while exposing only one shared read and one shared write service per pair and cycle?*

This paper makes four contributions:

- an equal-storage organization with three independent pairs per router, one private adaptive VC and one escape VC per direction, and four movable adaptive slots per pair;
- an alternating read schedule and a one-write admission guard that implement the shared bank as 1R1W;
- a conserved ownership-credit policy combining pressure, sticky ties, a three-credit cap, and idle-slot reclaim; and
- a 651-run evaluation that reports both the favorable region and the workloads where the fixed phase schedule is counterproductive.

2 Background and Motivation

A conventional four-VC input assigns four physical VC slots to each direction. Credit round-trip time determines how quickly a released downstream slot can be reused. With one-flit control traffic and an approximately 17-cycle credit loop at link latency $L = 8$, four slots impose an idealized zero-contention input-credit bound of

$$\lambda_{\text{credit}} \approx \frac{4}{17} = 0.235 \text{ flits/cycle.} \quad (1)$$

The exact saturation point can be lower because routing, output contention, and switch arbitration intervene.

TIDAL targets the case in which one member of an opposing pair repeatedly approaches this limit while the peer is underused. It does not attempt to replace the escape channel used by deadlock-free routing. Deterministic escape resources remain private, consistent with the separation of deadlock avoidance from adaptive resource use [4, 6].

3 Architecture

3.1 Direction-paired storage

Each router instantiates East/West, North/South, and Up/Down pairs. Figure 1 shows one pair. For each direction, RES contains one private adaptive VC and one private escape VC. The pair shares four adaptive slots:

$$2(1_{\text{private}} + 1_{\text{escape}}) + 4_{\text{pool}} = 8 \text{ slots.} \quad (2)$$

This equals the eight slots in two four-VC private inputs, so performance comparisons preserve total storage.

POOL membership follows *true ingress*, not the selected output. A flit entering from X may occupy the X-pair

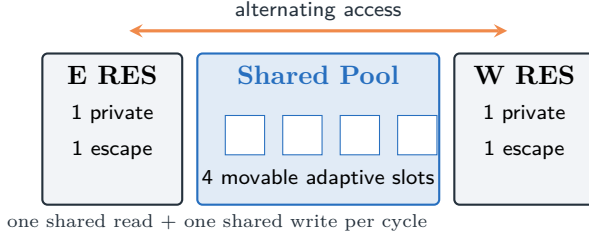


Figure 1: One direction pair. Every router contains three copies: E/W, N/S, and U/D. Capacity matches the private four-VC baseline.

POOL and later depart through Y or Z. Each pooled flit stores its ingress direction and upstream VC so that its credit returns to the correct sender.

3.2 Router-level on-chip organization

Figure 2 places one pair inside a router tile. The same structure is replicated for E/W, N/S, and U/D. The two incoming links retain independent RES banks. Their adaptive arrivals also feed one shared POOL write arbiter. The four POOL entries may be implemented as a compact central array or as two physical halves with equivalent global arbitration; the Garnet model specifies the externally visible service contract rather than an SRAM circuit.

3.3 One-read, one-write service

The POOL accepts at most one read and one write service per pair, control virtual network, and cycle. Read phases are complementary:

Phase	Side 0 (E/N/U)	Side 1 (W/S/D)
Even	RES	POOL
Odd	POOL	RES

Thus E can inspect RES while W inspects the POOL, and the roles reverse next cycle. Cycle parity structurally eliminates simultaneous cross-direction POOL reads. Ordinary VC and switch arbitration still select among eligible flits inside the active bank.

A per-pair write guard permits one adaptive POOL allocation each cycle. If both sides arrive, the second arrival may still use its private adaptive VC when that reserved credit is available; otherwise it waits. Escape storage never enters the shared bank and follows deterministic non-wrapping XYZ routing.

The port count must be stated precisely. A pair can perform two useful reads in a cycle: one from the selected side’s RES and one from the shared POOL. Only the latter consumes the Pool’s single read port. A RES write is likewise independent of the Pool write. For the present one-flit control traffic, Pool VC allocation and data arrival are one-to-one, so accepting at most one new Pool allocation per pair and cycle models a single Pool write service. A multi-flit extension would need a separate per-flit write schedule.

In a physical implementation, the phase is the low bit of the router cycle counter, the Pool writer is a two-request arbiter, and the Pool reader is a four-entry bank selector

Table 1: The two credit planes in TIDAL.

Plane	State and path	Meaning
Occupancy	Garnet OutputUnit VC state; returned on the real CreditLink to the upstream port that sent the flit	The physical slot is free; prevents overflow
Ownership	One owner bit per Pool slot; selected by the downstream pair controller and observed by either opposing sender	This direction may allocate the idle slot

followed by a true-ingress-lane multiplexer. The event-level Garnet implementation does not synthesize these blocks; it enforces their service limits and measures their utilization.

4 Pool-Credit Control

Each of the four shared slots has exactly one logical owner. For the E/W pair, ownership obeys

$$C_E + C_W = 4, \quad 1 \leq C_E, C_W \leq 3. \quad (3)$$

An owned POOL credit authorizes the corresponding upstream direction to reserve an idle slot. Private and escape credits remain pinned.

4.1 Two credit planes

The implementation deliberately separates *buffer occupancy* from *Pool ownership*. They are both called credits informally, but they have different destinations and correctness roles:

Returning an occupancy credit never changes its physical path. Even when an E flit was placed in the W-hosted half of the POOL, stored metadata identifies the true E ingress link and the original upstream logical VC. The ordinary free credit is sent back on E’s CreditLink. Independently, the downstream owner bit may stay with E or move to W. A sender whose logical VC has become idle must still pass the owner-table check before reusing that Pool slot.

4.2 Allocation, release, and handoff

Figure 3 follows one one-flit packet. The upstream router first searches its private adaptive VC, then idle Pool slots it already owns, then idle peer-owned slots that it may reclaim under the cap. A successful Pool choice atomically sets `owner[slot]`, sets `busy[slot]`, and consumes the pair’s write token for that cycle. The logical Pool VC ID travels with the flit. At the downstream router, that ID is mapped to one of four physical Pool entries and tagged with true ingress, credit input port, and upstream VC ID.

When the tail wins switch traversal, the downstream router first clears the busy bit and then evaluates the release policy. Table 2 gives the exact state transitions for the proposed pressure policy.

Pressure grant. When a pooled flit departs, its released credit is assigned to the side whose RES occupancy is

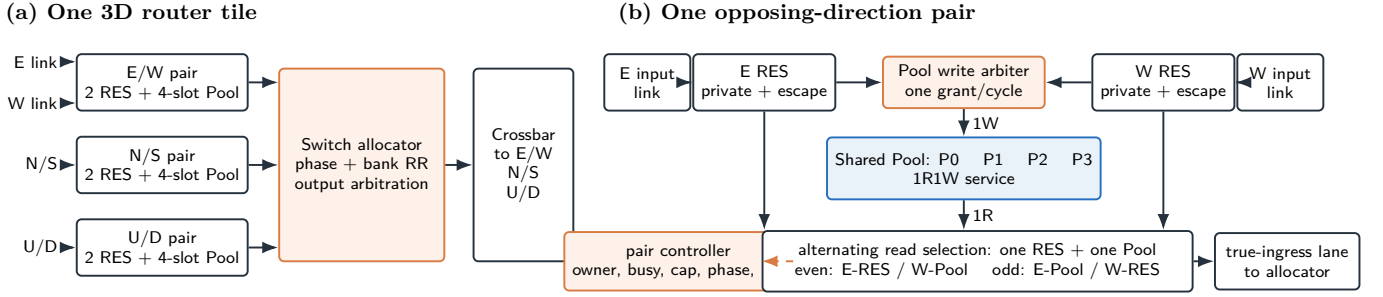
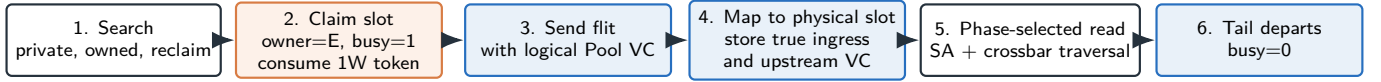


Figure 2: Proposed on-chip organization. (a) Three direction-pair controllers operate in parallel around the normal Garnet switch allocator and crossbar. (b) One pair exposes a single shared Pool read and write service. RES access remains separate; solid arrows are flit/request paths and the sole dashed arrow is controller state. The Pool result is placed on the flit’s true-ingress lane.

Allocation and forward path



After step 6: two independent feedback planes

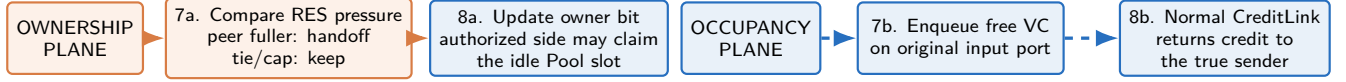


Figure 3: Pool-credit lifecycle. The ordinary free credit returns to the actual sender, whereas the released slot’s ownership may be granted to either side. The two non-crossing feedback lanes are both triggered by step 6. The simulator models the owner-table change with no additional grant delay.

larger. This local pressure signal approximates near-term need without adding a traffic predictor.

Sticky tie. Equal pressure preserves the previous owner. Stickiness avoids credit oscillation and favors the side likely to continue a burst.

Owner cap. A cap of three lets a hot side borrow one additional slot while guaranteeing one shared credit to its peer. Cap four is evaluated only as a full-elasticity upper bound.

On-demand reclaim. Release-only movement can freeze if an idle owner has no flit to release. A requester below its cap can therefore take an *idle* peer-owned slot immediately. Busy slots are never revoked. In the current simulator, this owner-table update adds no cycle; Section 8 treats this as an optimistic timing assumption.

The pressure value in the current code is the number of active private adaptive and escape VCs in RES. The just-released Pool slot is excluded. On equal pressure the old owner keeps the slot. If the preferred peer already owns three slots, the cap overrides pressure and the old owner keeps it.

4.3 Timing interpretation

The ordinary occupancy credit is not idealized: it is inserted into the original InputUnit’s credit queue and scheduled on Garnet’s CreditLink one router cycle later. Ownership handoff is the optimistic component. The simulator stores `owner[router][pair][vnet][slot]` at network scope, so the newly authorized upstream direction observes the new owner without a modeled sideband-wire

or grant-register delay. A physical design needs a one-way owner-grant signal from the downstream pair controller to both opposing upstream allocators. Adding a registered one-cycle grant, or a parameterized delay, is necessary before drawing frequency-sensitive conclusions.

Blind owner round-robin is a useful control policy, but it is distinct from the round-robin arbitration inside the read scheduler. Rotating ownership without observing demand may park capacity at an inactive side.

5 Garnet Implementation

We implemented TIDAL in gem5 Garnet’s router, input unit, VC allocator, and switch-allocation paths. The exact strict-1R1W snapshot is retained at `1R1W/overlay`; its provenance records implementation commit `dc43a290` and base commit `2eb6a4e`. Major additions are:

- physical-to-logical mapping for pair-global POOL slots;
- true-ingress and upstream-VC metadata for routing and credit return;
- complementary RES/POOL eligibility in switch allocation;
- a one-write-per-pair guard in downstream allocation;
- owner tables, pressure grants, cap enforcement, sticky ties, and idle-slot reclaim; and
- structural counters and X-biased/reversal synthetic traffic.

Table 2: Exact allocation and return rules for direction d and peer $\bar{d}$. An owner change is legal only for an idle slot.

Event	Guard / priority	Atomic state update	Result
Private allocation	Private adaptive VC is idle	Reserve the private VC; Pool state unchanged	No Pool port consumed
Owned Pool allocation	Private busy; Pool 1W token free; an idle slot has owner[s] = d	busy[s] $\leftarrow$ 1; record current-cycle write token	Send using logical VC 1 + s
Demand reclaim	No usable owned slot; 1W token free; peer-owned s is idle; $C_d < \text{cap}$	owner[s] $\leftarrow d$; busy[s] $\leftarrow$ 1; consume write token	Immediate reclaim; busy slots are never revoked
Second Pool request	Current-cycle write token already consumed	Do not allocate Pool; use private/escape only if independently legal	Otherwise request waits
Pool tail release	Selected Pool flit completes switch traversal	busy[s] $\leftarrow$ 0; count RES occupancy on both sides	Evaluate pressure below
Pressure handoff	RESused $_{\bar{d}} >$ RESused $_d$ and $C_{\bar{d}} < \text{cap}$	owner[s] $\leftarrow \bar{d}$	Peer gains ownership; old sender still receives the normal free credit
Sticky keep	Tie, own side fuller, or peer at cap	owner[s] $\leftarrow d$	Preserve burst locality and the one-credit peer floor

Table 3: VC numbering for the four-physical-VC design.

Namespace	Offset(s)	Role
Physical InputUnit	0	private adaptive
Physical InputUnit	1, 2	resident Pool half
Physical InputUnit	3	private escape
Internal-link logical	0	private adaptive
Internal-link logical	1–4	pair-global Pool slots
Internal-link logical	5	private escape

5.1 Physical and logical VC mapping

Each physical InputUnit still contains four control-vnet VC records. Offsets 0 and 3 are private; offsets 1 and 2 are that InputUnit’s physical half of the pair Pool. An internal link exposes six *logical* IDs so either opposing sender can name all four shared slots:

On arrival, `dpPhysMapLogicalVC` translates the pair-global slot ID into a physical input direction and offset. A borrowed E flit may therefore be written into an entry resident in the W InputUnit. The VC records true direction, original credit input port, and original upstream VC. Routing uses the true direction, not the physical host, and switch traversal uses E’s crossbar lane. On release, the saved fields steer the ordinary credit back to E.

5.2 Control state and pipeline hooks

Conceptually, each pair and control vnet needs four owner bits, four busy indications, one policy-RR bit, a current-cycle Pool-write token, and bank-RR pointers. Busy state can be derived from or colocated with existing VC-valid state in RTL. The global cycle parity supplies the read phase. Per-VC true-ingress and upstream-ID tags are required only because a logical Pool slot may be physically hosted by its peer. Statistics and ownership-wait timestamps in the C++ model are instrumentation, not proposed hardware.

The relevant Garnet hooks are:

- `Router::dpPhysSelectAdaptiveVC`: private first, then owned Pool, then idle peer Pool under the cap;
- `InputUnit::wakeup`: logical-to-physical redirection and ingress metadata capture;
- `SwitchAllocator::arbitrate_dp_phys_inports`: complementary RES/Pool eligibility and per-bank round-robin selection;
- `InputUnit::increment_credit`: true-link credit return and, on the free signal, release-time owner selection; and
- `GarnetNetwork::dpPhysClaimSlot`: owner/busy update and the one-write-per-pair-per-cycle guard.

Experiments inject only virtual network 0 with one-flit control packets and one slot per control VC. Hence $L = 8$ means an eight-cycle link, not an eight-flit packet. Exported counters include POOL allocations, owned allocations, demand reclaims, release handoffs/keeps, owner migrations, write blocks, bank reads, per-direction activity, and peak borrowing.

6 Evaluation

6.1 Methodology

The primary baseline, PRIVATE-V4, dedicates four physical VCs to every input. All TIDAL arms use two RES slots per side plus four shared slots per pair, preserving eight slots. We compare a non-migrating round-trip-static policy (RTS), blind owner round-robin, pressure cap three, and pressure cap four.

Tests use $4 \times 4 \times 4$ or $8 \times 4 \times 4$ 3D torus networks, link latencies $L \in \{1, 8\}$, one-flit control packets, 10,000 simulated cycles for main long runs, and three seeds per reported point. The 651-run main matrix contains 120 broad validation, 96 policy, 72 owner-cap, 192 reversal, and 171 X-biased/VC-scaling runs. A separate 48-run

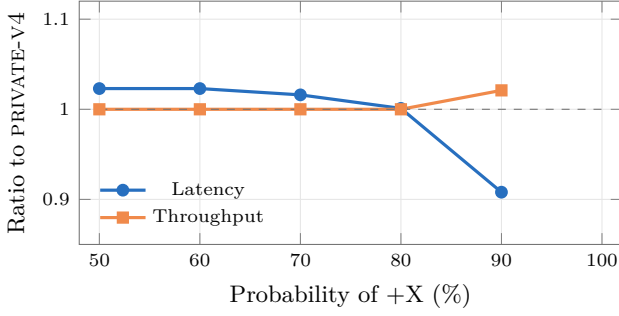


Figure 4: PRESSURE-CAP3 under X-direction skew at $L = 8$, injection 0.10. The fully skewed saturated point is excluded from the latency curve.

recovery-instrumentation rerun is not included in this total.

6.2 Structural validation

All 651 main runs completed without deadlock. Instrumented runs maintained credit conservation, never exceeded the owner cap, and satisfied

$$N_{\text{alloc}} = N_{\text{owned}} + N_{\text{reclaim}}, \quad (4)$$

$$N_{\text{migration}} = N_{\text{reclaim}} + N_{\text{handoff}}. \quad (5)$$

Per-direction allocation totals matched aggregate counts, and observed cross-direction POOL read conflicts were zero. Under cap three, peak borrowing was one slot, confirming that the peer retained its final credit.

6.3 Persistent X-direction skew

The $8 \times 4 \times 4$ X-biased workload sends each packet three minimal X hops and varies the probability of choosing +X. It preserves spatially distributed destinations while controlling directional imbalance. Figure 4 reports $L = 8$, injection 0.10, and four total VCs per input in the baseline.

At 50/50 and 60/40, TIDAL pays a 2.3% latency penalty because little cold-side capacity is stranded. The penalty falls to 1.6% at 70/30 and approximately breaks even at 80/20. At 90/10, latency falls by 9.2% while throughput rises by 2.1%. At 100% +X, the private baseline saturates, so a latency ratio is not a valid below-saturation comparison; the meaningful observation is 1.818 times delivered throughput during the measurement window.

Using completion/acceptance near one as a coarse criterion, the highest tested stable offered rates are 0.08 for PRIVATE-V4, 0.10 for cap three, and 0.12 for cap four. Cap three therefore extends the tested stable bracket by 25%; cap four provides an optimistic 50% upper bound. These are discrete brackets, not precise saturation thresholds.

6.4 Tornado and policy comparison

On $4 \times 4 \times 4$, torus tornado advances each packet by +1 in X, +1 in Y, and +1 in Z, making W, S, and D true inputs persistently hot. Table 7 shows $L = 8$ and injection 0.9.

Pressure cap three lowers latency by 4.85% without changing throughput materially because an output bottleneck limits tornado first. It captures about 83% of the latency benefit of cap four while preserving one peer credit. Only approximately 0.49% of POOL allocations use on-

demand reclaim: stable ownership handles most demand, and reclaim resolves occasional mismatches. Owner round-robin loses 56.5% throughput because blind rotation can place credits at an inactive peer and has no demand-driven correction.

Cap two prevents borrowing and is behaviorally equivalent to static ownership. Cap three recovers most of cap four’s latency benefit while retaining one credit at the cold side. Cap four is slightly faster here but can temporarily leave the peer with no Pool ownership.

The event identities are visible numerically: $202,267 = 201,267 + 1,000$ allocations and $1,809 = 1,000 + 809$ migrations. Completed Pool reads also equal released keeps plus handoffs, $202,083 = 201,274 + 809$; the allocation/read difference is the set of flits still resident at measurement end. Zero write-block counts mean the allocator never committed a second Pool write in the same pair-cycle in these runs; this validates the guard’s behavior, not the timing or area of a fabricated 1W array.

6.5 Where 1R1W loses

The phase schedule is a real bandwidth constraint. Against PRIVATE-V4, uniform traffic at $L = 8$, injection 0.9 raises latency by approximately 3.2%. X-opposite traffic raises latency by 3–15% because both members of a pair compete for one shared read. Tornado at $L = 1$ raises latency by 25.1% because the credit loop is too short for additional capacity to repay the phase cost. Reversing the hot X direction every 16 cycles raises latency by 7.8% while ownership and old traffic drain in opposite directions.

The epoch-16 point is worst because ownership moves most often while the previous direction’s flits are still draining. Very fast four-cycle reversal leaves little time to build an extreme ownership split; long epochs amortize each transition. This is direct evidence that credit policy must be evaluated as a dynamic controller, not only by its steady-state cap.

VC scaling supports the same interpretation. Near the measured knee, a six-VC organization reaches a 1.27 throughput ratio and a 0.56 latency ratio, extending the stable bracket from 0.12 to 0.15. At eight VCs, where private capacity already hides more of the credit loop, the ratios shrink to 1.08 and 0.83, extending 0.18 to 0.20.

6.6 Data provenance

The raw rows used above are tracked under `1R1W/overlay/lab/DP/physical_bound/`. Table 4 names each CSV exactly. `run_sweep_dpphys.py` records the gem5 command for every run, and `analyze_dpphys.py` performs paired-by-seed ratios and confidence-interval aggregation. The archived overlay should be reconstructed in a temporary worktree from base commit `2eb6a4e`; it should not be copied over the repository’s current working source.

7 Hardware Interpretation

The evaluated comparison establishes equal buffer capacity and a one-read, one-write POOL service limit. This is a plausible manufacturing advantage over an ideal shared bank that must serve both directions simultaneously, but

Table 4: Complete 651-run evaluation matrix. Rates are offered injection probabilities. Long-run rows report three-seed means and approximate 95% confidence intervals; validation is a single-seed smoke matrix.

CSV dataset	Purpose / traffic	Topology	L	Sweep points	Cycles $\times$ seeds	Runs
validation_v4	Sanity: uniform, X-opposite, tornado	$4 \times 4 \times 4$	1, 8	rate 0.1, 0.3, 0.5, 0.7, 0.9; four modes	$3,000 \times 1$	120
policy_v4	Tornado policy comparison	$4 \times 4 \times 4$	1, 8	rate 0.1, 0.3, 0.5, 0.9; four modes	$10,000 \times 3$	96
cap_v4	Tornado/uniform cap ablation	$4 \times 4 \times 4$	8	rate 0.3, 0.5, 0.9; caps 2, 3, 4 + private	$10,000 \times 3$	72
reversal_v4	Alternating $+X/-X$ demand	$4 \times 4 \times 4$	1, 8	rate 0.3, 0.9; epoch 4, 16, 64, 256; four modes	$10,000 \times 3$	192
xbiased_bias_v4	Persistent X-direction skew	$8 \times 4 \times 4$	8	rate 0.10; $+X$ bias 50–100%; private/cap3/cap4	$10,000 \times 3$	54
xbiased_fullskew_v4	Four-VC stability bracket	$8 \times 4 \times 4$	8	rate 0.08–0.12; 100% $+X$; three modes	$10,000 \times 3$	45
xbiased_vcscale	Six/eight-VC scaling	$8 \times 4 \times 4$	8	rate 0.15, 0.18, 0.20, 0.25; six modes	$10,000 \times 3$	72
Total						651

Table 5: Detailed X-bias sweep at $L = 8$, offered rate 0.10. Values are mean $\pm$ approximate 95% confidence interval across three seeds; ratios are computed per seed before averaging. The dagger marks an above-saturation latency comparison.

$+X$	Private thr.	Cap3 thr.	Thr./base	Private net lat.	Cap3 net lat.	Cap3 lat./base	Cap4 lat./base
50%	.0496 $\pm$.0004	.0496 $\pm$.0004	1.000 $\pm$.000	45.86 $\pm$.02	46.94 $\pm$.02	1.023 $\pm$.000	1.018 $\pm$.000
60%	.0496 $\pm$.0004	.0496 $\pm$.0004	1.000 $\pm$.000	46.05 $\pm$.06	47.09 $\pm$.04	1.023 $\pm$.001	1.015 $\pm$.001
70%	.0496 $\pm$.0004	.0496 $\pm$.0004	1.000 $\pm$.000	46.65 $\pm$.08	47.39 $\pm$.02	1.016 $\pm$.002	1.005 $\pm$.002
80%	.0496 $\pm$.0004	.0496 $\pm$.0004	1.000 $\pm$.000	47.91 $\pm$.15	47.93 $\pm$.07	1.001 $\pm$.002	.982 $\pm$.002
90%	.0486 $\pm$.0013	.0496 $\pm$.0004	1.021 $\pm$.028	53.94 $\pm$ 4.68	48.78 $\pm$.07	.908 $\pm$.075	.880 $\pm$.074
100%	.0273 $\pm$.0008	.0496 $\pm$.0004	1.818 $\pm$.061	180.14 $\pm$ 8.17	50.30 $\pm$.24	.280 $\pm$.012 [†]	.265 $\pm$.012 [†]

it is not yet an area or energy result. TIDAL adds owner bits, physical-slot mapping, selectors, arbitration, control wiring, and a potentially critical ownership update path. We therefore claim reduced shared-port demand, not zero hardware cost. RTL synthesis against private and dual-access pool implementations is required to quantify area, energy, and frequency.

This comparison explains both the intended cost saving and the performance losses. Relative to an unrestricted 2R2W shared pool, TIDAL removes one shared read and one shared write service, reducing the hardest shared-port requirement. Relative to private banks, however, it adds a central selection and ownership path. Whether the smaller Pool port count outweighs those selectors depends on entry width, SRAM/register-file implementation, wire length, and target frequency. Garnet cannot answer that manufacturing question. The ordinary router allocator and crossbar remain present in all three organizations [5].

8 Limitations and Next Steps

The present evidence is preliminary and simulator-level. First, demand reclaim updates ownership with zero additional cycles; a one-cycle and parameterized grant path should measure timing sensitivity. Second, finer 0.005 injection steps, longer drain-aware runs, and at least five seeds are needed around every saturation knee. Third, a directed simultaneous E/W-write microbenchmark should stress the write guard. Fourth, multi-flit packets require an explicit packet/VC ownership-retention rule. Finally, RTL synthesis should compare private buffers, 1R1W TIDAL, and an idealized dual-access pool.

9 Related Work

Virtual-channel flow control and its buffer/throughput tradeoffs were formalized by Dally [3]. Deadlock-free routing can be constructed through channel-dependency restrictions [4] or broader adaptive conditions [6]. TIDAL preserves a private deterministic escape resource rather than changing those guarantees. Garnet provides the detailed router and credit model used here [1]; gem5 provides the surrounding simulation infrastructure [2]. Our contribution is specifically the combination of direction-paired equal-capacity storage, a structurally 1R1W shared bank, and demand-aware ownership credits.

10 Division of Labor

The project was completed with equal overall contribution: Changxun Pan 50% and Liming Wang 50%. Both contributors participated in architecture discussion, debugging, correctness review, experiment review, interpretation, writing, slide production, and presentation preparation.

Liming Wang proposed the original direction-paired pooling concept, implemented and evaluated the earlier 1R2W organization, expanded traffic and evaluation coverage, reviewed the 1R1W and credit-policy behavior, and co-analyzed and co-authored the final deliverables.

Changxun Pan co-developed the design and finalized its complementary 1R1W phase schedule, designed pressure, sticky-tie, cap-three, and reclaim control, implemented the final mechanism and instrumentation, ran and cross-checked major 1R1W experiments, and co-analyzed and co-authored the final deliverables.

Table 6: Full-skew stability sweep at $L = 8$. Each cell is injection acceptance / delivered throughput / network latency, averaged over three seeds. A design is treated as stably accepting the offered load only when acceptance remains close to one.

Offered rate	PRIVATE-V4		Pressure cap 3		Pressure cap 4	
0.08	.996	.0397 / 48.8	.996	.0397 / 48.2	.996	.0397 / 46.9
0.09	.932	.0416 / 64.3	.996	.0446 / 49.0	.996	.0446 / 47.3
0.10	.559	.0273 / 180.1	.997	.0496 / 50.3	.997	.0496 / 47.7
0.11	.391	.0208 / 271.9	.901	.0491 / 63.5	.997	.0545 / 48.3
0.12	.327	.0190 / 307.5	.484	.0283 / 154.2	.997	.0595 / 49.2

Table 7: Tornado policy comparison at offered rate 0.9. Values are mean $\pm$ approximate 95% confidence interval over three seeds. Pool, reclaim, and handoff columns are percentages of completed control events.

L	Mode	Throughput	Thr./base	Net latency	Lat./base	Pool service	Reclaim	Handoff
1	PRIVATE-V4	.4489 $\pm$.0003	1.000	10.46 $\pm$.00	1.000	—	—	—
1	RTS	.4488 $\pm$.0003	1.000 $\pm$.000	12.84 $\pm$.01	1.227 $\pm$.000	53.45	.00	.00
1	Owner-RR	.3138 $\pm$.0013	.699 $\pm$.003	32.08 $\pm$.16	3.067 $\pm$.016	25.37	.00	.11
1	Pressure cap3	.4488 $\pm$.0003	1.000 $\pm$.000	13.09 $\pm$.01	1.251 $\pm$.001	59.11	.13	.09
8	PRIVATE-V4	.1564 $\pm$.0000	1.000	49.37 $\pm$.07	1.000	—	—	—
8	RTS	.1554 $\pm$.0000	.994 $\pm$.000	58.25 $\pm$.05	1.180 $\pm$.001	51.35	.00	.00
8	Owner-RR	.0680 $\pm$.0004	.435 $\pm$.003	157.03 $\pm$ 1.16	3.181 $\pm$.022	27.67	.00	.44
8	Pressure cap3	.1563 $\pm$.0000	1.000 $\pm$.000	46.97 $\pm$.03	.952 $\pm$.002	66.38	.49	.40

Table 8: Owner-cap ablation for tornado, $L = 8$, rate 0.9.

Mode	Lat.	Lat./base	Reclaim	Borrow peak
PRIVATE-V4	49.37	1.000	—	0
Cap 2	58.25	1.180	0.00%	0
Cap 3	46.97	.952	.49%	1
Cap 4	46.47	.941	.19%	2

Table 9: Credit/control event counts for pressure cap 3 under tornado, $L = 8$, rate 0.9. Counts are rounded three-seed means per 10,000-cycle run.

Counter	Mean	Counter	Mean
Pool allocations	202,267	Owned allocations	201,267
Demand reclaims	1,000	Owner migrations	1,809
Release handoffs	809	Release keeps	201,274
Pool reads	202,083	RES reads	102,347
Pool write blocks	0	Pool read conflicts	0
Borrowed peak	1	Deadlocks	0

11 Conclusion

TIDAL shows that buffer elasticity does not require an unconstrained global pool. Pairing opposite directions and preserving private escape capacity lets a four-slot adaptive pool move storage toward sustained pressure while retaining equal total capacity. A complementary schedule enforces one shared read and one shared write per cycle. The design is effective under persistent directional skew and long credit loops, but balanced demand, short links, and fast reversals expose its serialized-access cost. Pressure-aware credit movement is essential: blind ownership rotation is substantially worse. The next decisive step is to add grant delay and synthesize the control and storage organization so that performance gains can be judged against measured area, energy, and timing.

References

- [1] N. Agarwal, T. Krishna, L.-S. Peh, and N. K. Jha. GAR-NET: A detailed on-chip network model inside a full-system

Table 10: Representative pressure-cap3 losses. Ratios are against equal-storage PRIVATE-V4. The X-opposite row is a 3,000-cycle single-seed smoke point; the others are 10,000-cycle, three-seed results.

Traffic / condition	L	Thr./base	Lat./base
Uniform, rate .9	8	1.001	1.032
X-opposite, rate .9	8	.999	1.154
Tornado, rate .9	1	1.000	1.251
X reversal, epoch 16	8	.999	1.078

Table 11: Direction-reversal sensitivity at $L = 8$, rate 0.9. Counts and absolute values are three-seed means.

Epoch	Base thr.	Cap3 thr.	Base lat.	Cap3 lat.	Migrations
4	.1575	.1575	27.37	27.97	5,335
16	.1560	.1559	28.49	30.72	16,707
64	.1549	.1558	27.77	28.60	7,649
256	.1565	.1567	26.63	27.26	2,428

simulator. In *Proceedings of the IEEE International Symposium on Performance Analysis of Systems and Software*, pages 33–42, 2009.

- [2] N. Binkert, B. Beckmann, G. Black, S. K. Reinhardt, A. Saidi, A. Basu, J. Hestness, D. R. Hower, T. Krishna, S. Sardashti, R. Sen, K. Sewell, M. Shoaib, N. Vaish, M. D. Hill, and D. A. Wood. The gem5 simulator. *ACM SIGARCH Computer Architecture News*, 39(2):1–7, 2011.
- [3] W. J. Dally. Virtual-channel flow control. *IEEE Transactions on Parallel and Distributed Systems*, 3(2):194–205, 1992.
- [4] W. J. Dally and C. L. Seitz. Deadlock-free message routing in multiprocessor interconnection networks. *IEEE Transactions on Computers*, C-36(5):547–553, 1987.
- [5] W. J. Dally and B. Towles. *Principles and Practices of Interconnection Networks*. Morgan Kaufmann, 2004.
- [6] J. Duato. A new theory of deadlock-free adaptive routing in wormhole networks. *IEEE Transactions on Parallel and Distributed Systems*, 4(12):1320–1331, 1993.